RPC-Bench: A Fine-grained Benchmark for Research Paper Comprehension

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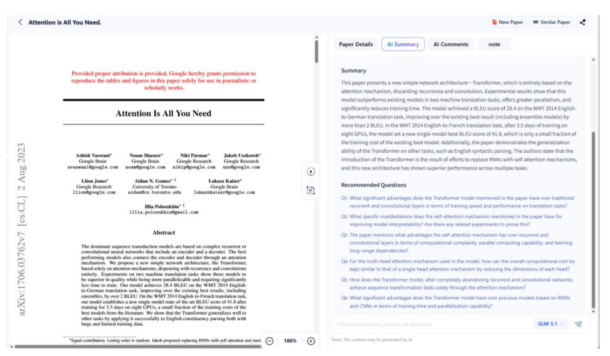
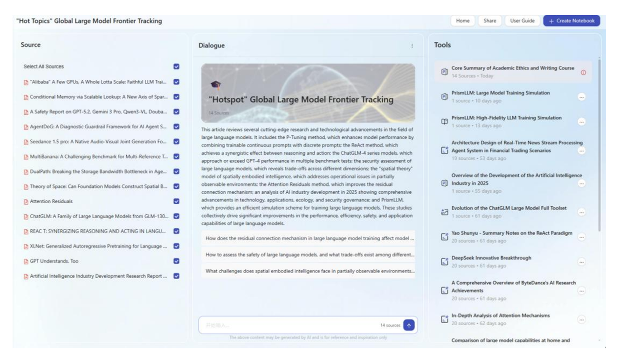
Project Page: https://rpc-bench.github.io/

Why Research Paper Comprehension?

Research paper comprehension is a key prerequisite for **reliable research copilots** and **subsequent scientific discovery**.



AI Paper Reading



Research Labs

Related Benchmarks

- Limited scale and paper coverage
- Rely on synthetic QA pairs
- Lack fine-grained comprehension taxonomy
- Provide narrow evaluation metrics
- Do not jointly support text and visual inputs

Benchmarks	Papers	QA	Real QA	Taxonomy	Eval. Metrics	Textual inp.	Visual inp.
PeerQA	208	579	✓	task	Corr.	✓	✗
SPIQA	25.5K	270K	✗	task	LLMLogScore	✓	✓
ArXivQA	16.6K	-	✗	task	-	✗	✓
DocGenome	500K	-	✗	task	GPT-acc	✗	✓
RPC-Bench	4150	61.3K	✓	content	Conc., F1-like	✓	✓

Table 1: Comparison with relevant research paper Benchmarks. Conc.=Conciseness; Corr.=Correctness; F1-like is defined as the harmonic mean of correctness and completeness; inp.=input. "Eval. Metrics" are LLM-based metrics.

Evaluation Protocol

- Uses accuracy for binary claim verification and LLM-as-a-Judge scoring for open-ended QA.
- Scores correctness, completeness, and conciseness on a 0-5 scale; derives F1-like from correctness-completeness and informativeness by penalizing verbose answers.

$$\text{F1-like} = \frac{(1+\beta^2) \times (\text{Correctness} \times \text{Completeness})}{\beta^2 \times \text{Correctness} + \text{Completeness}}$$

$$\text{Informativeness} = \text{F1-like} \times \frac{\text{Conciseness}}{5}$$

Adopts the most human-aligned setup: **title/abstract-augmented prompts, separate dimension scoring, decimal scores, and averaged GPT-5/Gemini-3-pro judgments**, selected by Bradley-Terry correlation and pairwise AUC

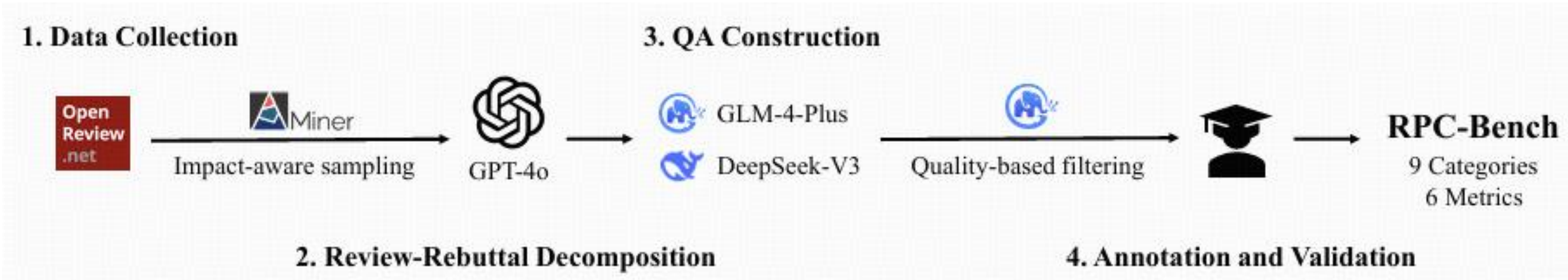
Setting	P-BT	S-BT	PW-AUC	Avg.
AUG+SEP	0.8955	0.9137	0.7125	**0.8406**
AUG+JOI	0.9003	0.9091	0.6966	<u>0.8353</u>
RAW+SEP	0.8773	0.9137	0.7054	0.8321
RAW+JOI	0.8759	0.9137	0.7100	0.8332

Table 4: Agreement between human and model judgment w.r.t. prompt configurations. (AUG: enhancement with title and abstract; SEP/JOI: separate/joint evaluation)

Model	P-BT	S-BT	PW-AUC	Avg.
GPT-5_{int}	0.9059	0.9227	0.7027	<u>0.8437</u>
Claude-4.5_{int}	0.8928	0.9136	0.7038	0.8367
Gemini-3_{int}	0.9091	0.9227	0.7090	**0.8469**
GPT-5_{dec}	0.9213	0.9137	0.7255	<u>0.8535</u>
Claude-4.5_{dec}	0.8873	0.9091	0.7197	0.8387
Gemini-3_{dec}	0.9170	0.9182	0.7330	**0.8561**

Table 5: Agreement between different LLM judges and human assessments. Subscript *int* denotes integer scoring, and subscript *dec* denotes decimal scoring.

RPC-Bench: Construction Pipeline



- Built from real review-rebuttal interactions through decomposition, rewriting, filtering, and human verification.
- Each paper is provided in structured-text and rendered-page formats for LLM/VLM evaluation.

Taxonomy Design

- Designed around the natural research flow of academic papers, progressing from what and how to why questions to trace the full logic of a paper.
- Includes free-form QA on concepts, methods, and experiments, plus binary claim verification for judging whether paper-grounded claims are true or false.

Task Taxonomy

1. Concept Understanding (C.U.) [What-4.27%]: Clarifies or explains key concepts, terminology, theoretical viewpoints, or information conveyed in figures, tables, or formulas.
2. Methods
   - 2.1. Method Disambiguation (M.D.) [What-8.91%]: Clarifies methodological details to resolve misunderstandings or ambiguities, ensuring an accurate understanding of proposed approaches.
   - 2.2. Method Mechanics (M.M.) [How-9.91%]: Questions about the implementation or function of methodological workflow or components, such as the effect of specific modules in models.
   - 2.3. Motivation Analysis (M.A.) [Why-6.29%]: Explores the rationale behind a proposed method or decision.
   - 2.4. Method Comparison (M.C.) [11.75%]: Compares the proposed approach with baseline methods, analyzing similarities, differences, or performance to highlight novelty.
3. Experiments
   - 3.1. Experimental Exposition (E.E.) [What-13.07%]: Explains results, infers how changes to experiments could affect results or conclusions, and handling reasoning tasks (i.e., calculations, counting, or comparisons).
   - 3.2. Experimental Setup (E.S.) [How-6.77%]: About the design, configuration, and execution of experiments.
   - 3.3. Experimental Analysis (E.A.) [Why-14.08%]: Studies the reasons of specific experimental results, links them to the proposed approach, and assesses their generalizability and potential impact.
4. Claim Verification (C.V.) [24.95%]: Binary classification tasks that judge whether claims, hypotheses, or experimental conclusions are correct.

RPC-Bench Overview

Statistics	train	val	test
Papers	3100	850	200
Accept	1980	609	116
year	2013-2021	2022-2023	2024
Venue	12	7	4
QA	45651	12895	2787
A/M Q	25.2/105	26.2/261	24.2/250
A/M A	70.5/297	121.5/1337	87.9/773

Table 2: Statistics of the RPC-Bench. A/M Q: average/max question length. A/M A: average/max answer length. Lengths are measured in words.

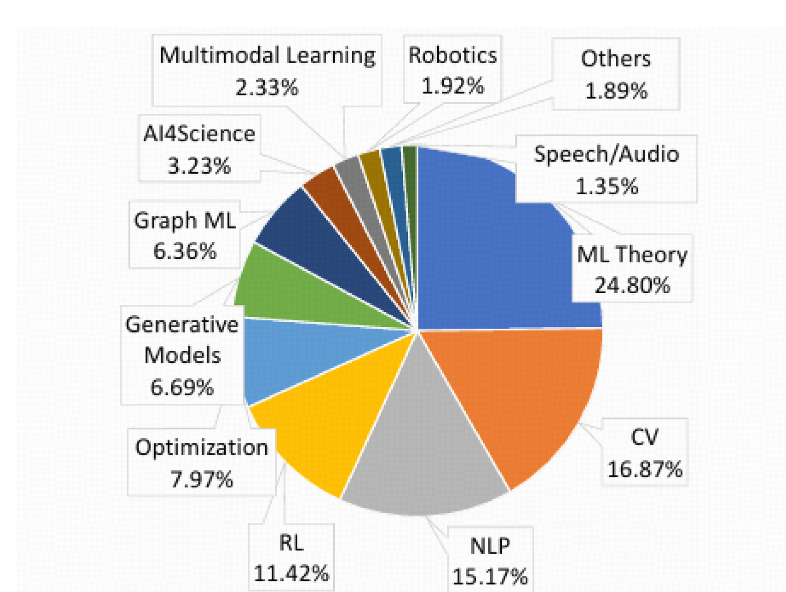


Figure 2: Domain distribution of RPC-Bench. ML: Machine Learning; CV: Computer Vision; NLP: Natural Language Processing; RL: Reinforcement Learning.

Experimental Results

- Traditional metrics fail to reflect true paper comprehension; correctness, completeness, and conciseness reveal clear gaps even for top models.
- Text-only LLMs outperform image-based VLMs, while small DCM/RAG models struggle with long-context reasoning.
- Models handle concept/method questions better than experimental analysis and claim verification, which require evidence integration and strict factual judgment.

Model Type	Model	Traditional: R-L	Traditional: BERTS.	LLM-as-a-judge: Concise.	LLM-as-a-judge: Correct.	LLM-as-a-judge: Complete.	LLM-as-a-judge: F1-like	LLM-as-a-judge: Info.
LLM	DeepSeek-V3.2	19.22	<u>55.60</u>	56.31	58.73	55.19	56.91	32.04
	GLM-4.7	17.09	48.58	54.34	54.36	51.75	53.02	28.81
	Qwen3	16.16	54.25	41.44	55.88	56.64	56.26	23.31
	GPT-5	16.89	54.52	54.93	**69.10**	**67.33**	**68.20**	**37.46**
	Claude-4	16.60	54.02	41.37	58.53	58.44	58.48	24.19
	Gemini-2.5	18.24	**55.67**	54.87	<u>62.65</u>	<u>59.03</u>	<u>60.79</u>	33.35
DCM	DocOwl2(V)	14.32	46.42	50.19	11.75	6.66	8.50	4.27
	Docopilot(V)	16.92	53.82	39.31	18.31	17.12	17.69	6.96
	Monkey(V)	**20.16**	55.19	54.61	17.08	11.27	13.58	7.41
VLM	GLM-4.6V	<u>19.38</u>	54.76	**64.55**	47.32	43.43	45.29	29.23
	Qwen3(V)	14.70	53.72	22.64	20.17	20.14	20.16	4.56
	GPT-5(V)	17.32	54.85	61.47	58.90	55.34	57.07	<u>35.08</u>
	Claude-4(V)	13.33	50.63	31.63	54.16	53.32	53.74	16.99
	Gemini-2.5(V)	17.27	54.85	51.71	48.39	45.59	46.95	24.28
RAG	HippoRAG2	18.71	54.16	45.77	33.13	27.88	30.28	13.86
	MemoRAG	13.55	52.70	51.31	24.19	19.10	21.35	10.96
	Raptor	18.35	54.00	36.47	25.28	20.82	22.84	8.33
	VdocRAG(V)	17.77	52.22	<u>61.54</u>	21.17	13.88	16.77	10.32
	VisRAG(V)	16.80	54.93	39.90	26.24	23.63	24.87	9.92

Table 3: Evaluation results of free-form QA on the test set. R-L=ROUGE-L; BERTS.=BERTScore; Concise.=Conciseness.; Correct.=Correctness; Complete. = Completeness; Info. = Informativeness. The best results are highlighted in **bold**, and the second-best results are <u>underlined</u>.

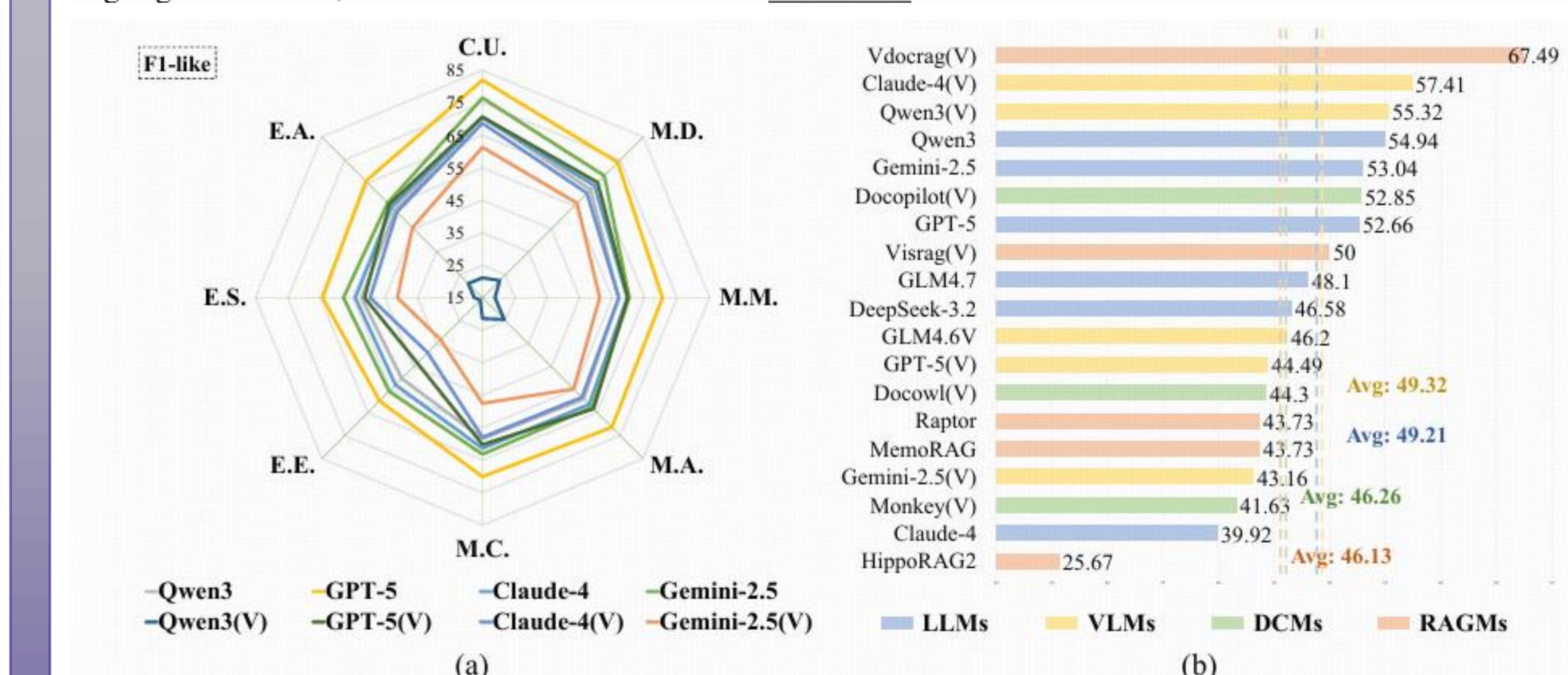


Figure 4: Comparison of LLMs and VLMs on open-ended question answering (F1-like score; left), and the performance of all models on claim verification tasks (ACC; right).

Conclusion

- Introduce RPC-Bench, a large-scale benchmark with 4,150 papers, 61.3K QA pairs, and 9 fine-grained categories aligned with the research workflow.
- Transform real review-rebuttal exchanges into high-quality QA pairs through an LLM-human collaborative annotation framework.
- Provide a comprehensive evaluation protocol that jointly measures correctness, completeness, and conciseness, with strong alignment to human judgments.
- Benchmark 28 advanced models across LLMs, VLMs, DCMs, and RAG systems in both text and page-image settings.